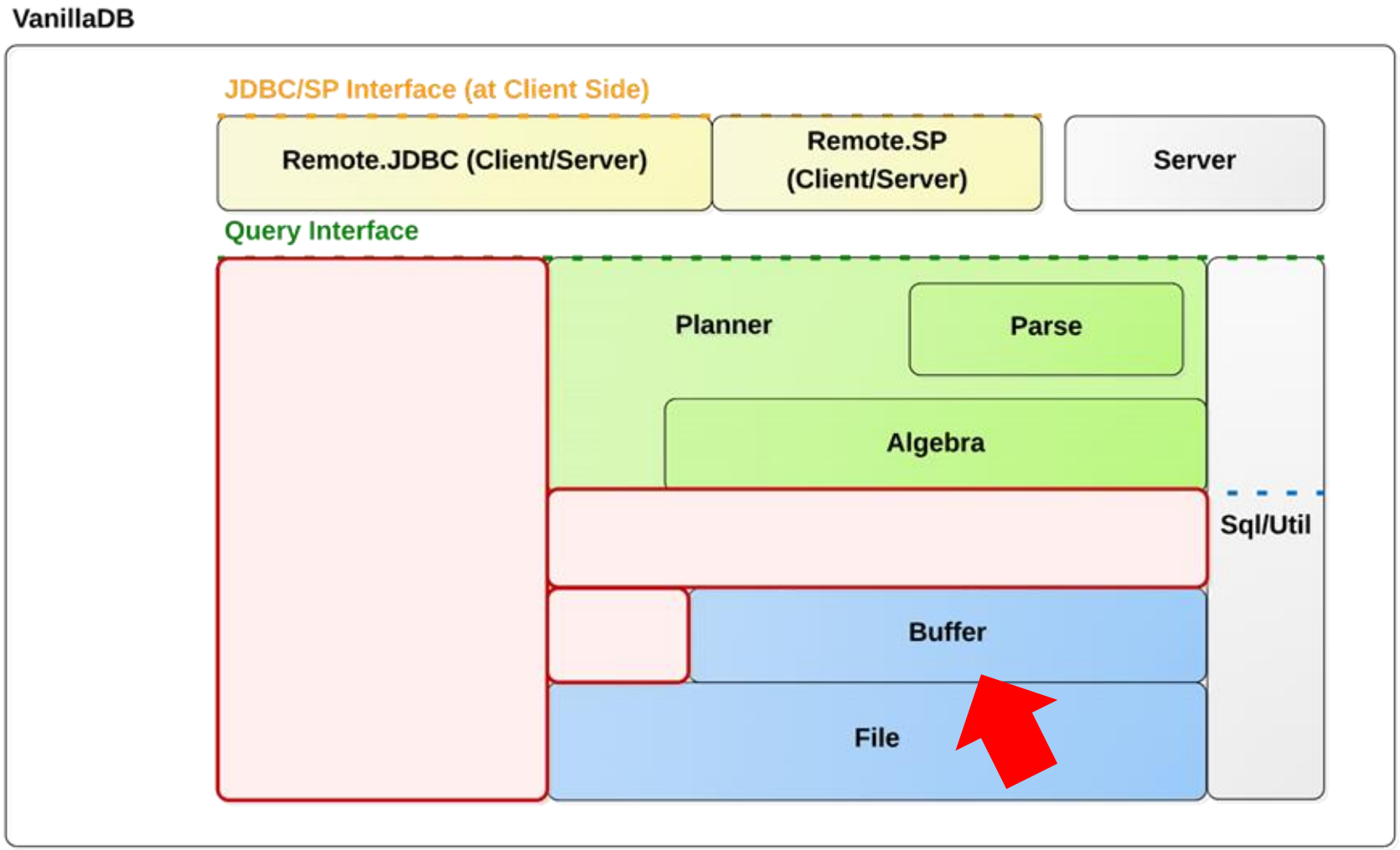
VanillaCore Walkthrough Part 4

Introduction to Database Systems

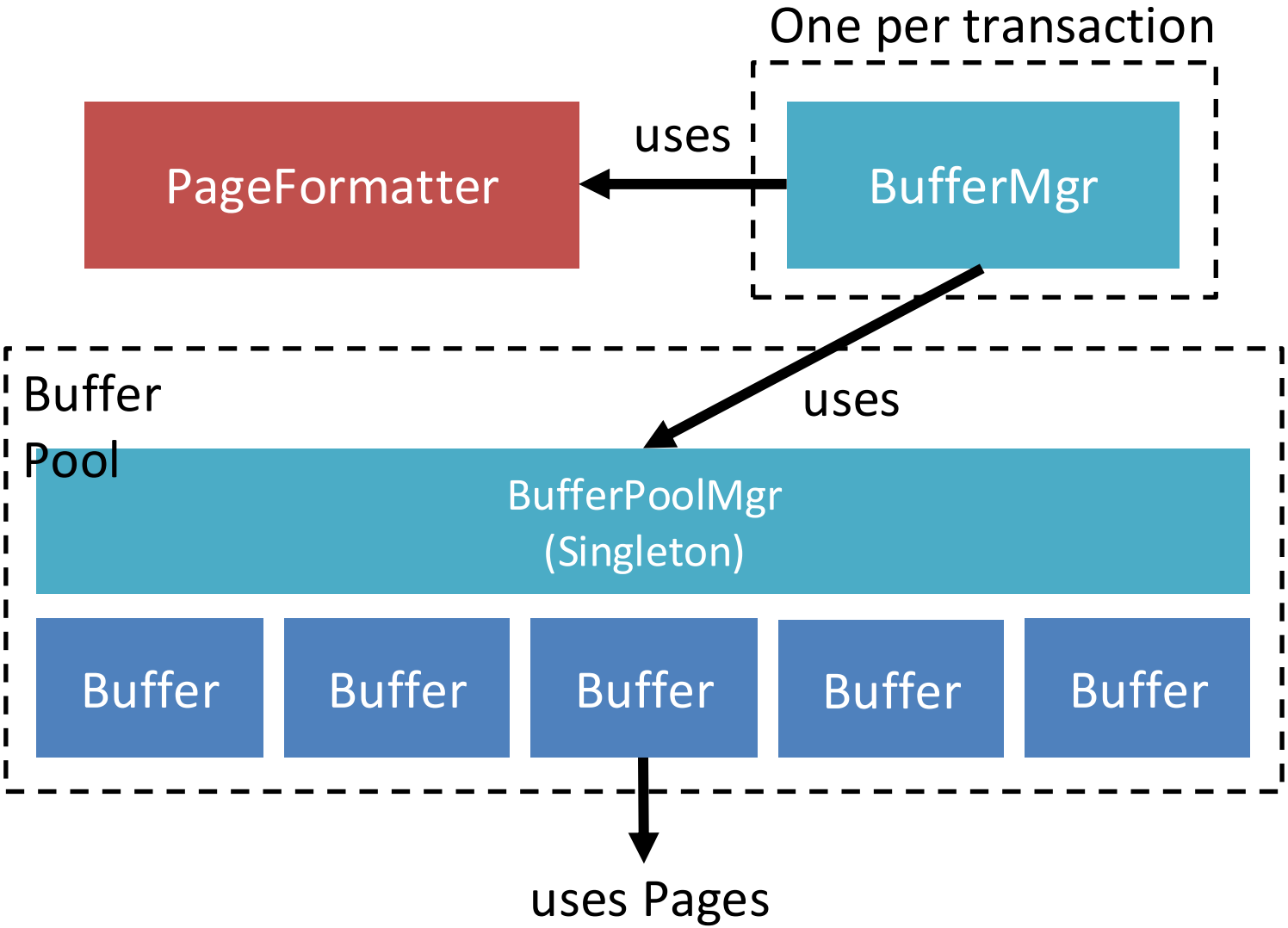
DataLab

CS, NTHU

Today's Focus



`buffer` Package



Functionality

- Main Components
  - `Buffer`: a memory space for caching a block
  - `BufferPoolMgr`: manages the buffer pool
  - `BufferMgr`: provides the access to buffers and manages the pinned buffers for each transaction
    - A transaction waits here if it could not pin any buffer.
  - `PageFormatter`: formats a fresh block

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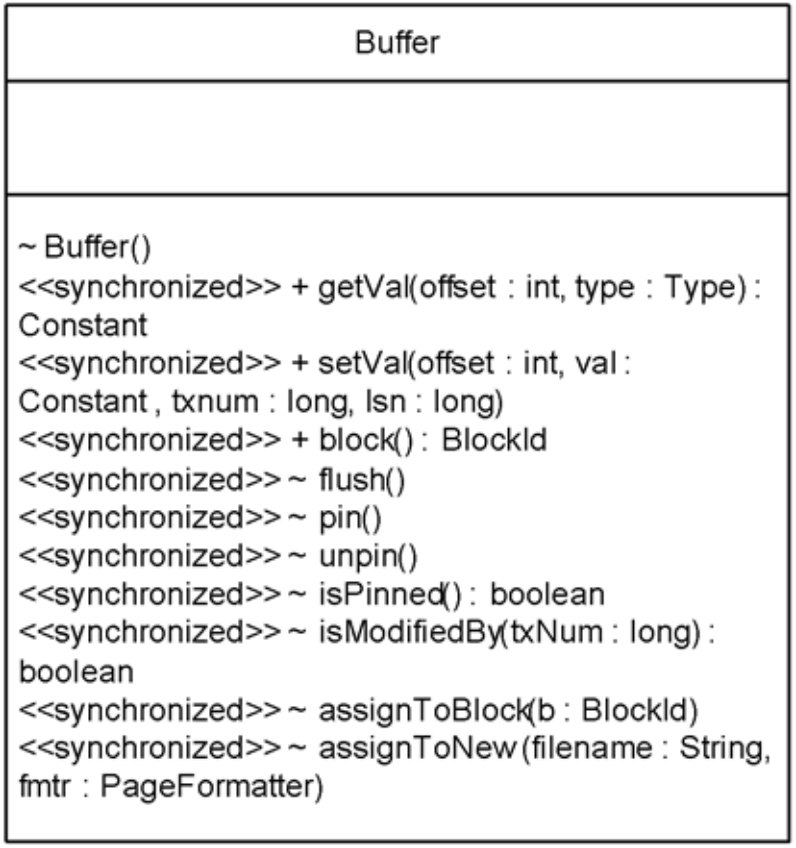
Buffer

- Wraps a page and stores
  - ID of the holding block
  - Pin count
  - Modified information
  - Log information
- ***Supports WAL***
  - `setVal()` requires an LSN
    - Must be preceded by `LogMgr.append()`
  - `flush()` calls `LogMgr.flush(maxLsn)`



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BufferPoolMgr

BufferPoolMgr
~ BufferPoolMgr(numbuffs : int) <<synchronized>> ~ flushAll() <<synchronized>> ~ flushAll(txnum : long) <<synchronized>> ~ pin(blk : BlockId) : boolean <<synchronized>> ~ pinNew(filename : String, fmtr : PageFormatter) : Buffer <<synchronized>> ~ unpin(buffs : Buffer[]) <<synchronized>> ~ available() : int

`BufferPoolMgr`

- Singleton
- Implement replacement strategy
- The `pin()` ***returns null immediately*** if there's no candidate buffer
  - Then, the `BufferMgr` make the calling thread waiting and retrying later

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BufferMgr

BufferMgr : TransactionLifecycleListener

<u><<final>> # BUFFER_POOL_SIZE : int</u>

+ BufferMgr()
+ onTxCommit(tx : Transaction)
+ onTxRollback(tx : Transaction)
+ onTxEndStatement(tx : Transaction)
+ pin(blk : BlockId)
+ pinNew(filename : String, fmtr : PageFormatter) : Buffer
+ unpin(buff : Buffer)
+ flushAll()
+ flushAll(txNum)
+ available() : int

`BufferMgr`

- Created when constructing a transaction
- A `BufferMgr` manages the pinned buffers and the pinning counts of a transaction
- `BufferMgr.pin()` makes the calling thread to wait if there's no candidate buffer for replacement

Java `wait()` and `notifyAll()` Methods

- In Java, every object has a waiting list
  - `obj.wait(timeout)` puts the caller thread into the waiting list of `obj`

- The thread will be removed from the list and ready for execution in two conditions:
  - Another thread call `obj.notifyAll()`
  - Timeout elapsed

Java `wait()` and `notifyAll()` Methods

- If…
  1. `obj.wait()` is surrounded by a synchronized block, and
  2. there are multiple threads in `obj`'s waiting list,
- Then when `notifyAll()` is called, ***all*** waiting threads will compete on the lock to enter the synchronized block
  - ***No*** FIFO guarantee which thread will be notified first, and which will acquire the lock first
  - Only one thread wins the lock, others ***blocked*** until the winner releases the lock

BufferMgr

- `pin()`: if `BufferPoolMgr` returns null, put the current thread into `BufferPoolMgr`'s waiting list

```
buff = bufferPool.pin(blk);
while (buff == null && !waitingTooLong(timestamp)) {
    bufferPool.wait(MAX_TIME);
    buff = bufferPool.pin(blk);
}
```

- `unpin(buff)`: notify all threads in `BufferPoolMgr`'s waiting list
  - Only one thread will pin successfully due to the synchronization

`BufferMgr` vs. `BufferPoolMgr`

- Each transaction has its own `BufferMgr`, but there is only one `BufferPoolMgr`

- Responsibility
  - `BufferPoolMgr` manages the buffer pool
  - `BufferMgr` handles waiting for pinning and manages pinned buffers for each transaction

```java
public Buffer pin(BlockId blk) {
    synchronized (bufferPool) {
        PinnedBuffer pinnedBuff = pinnedBuffers.get(blk);
        if (pinnedBuff != null) {
            pinnedBuff.pinnedCount++;
            return pinnedBuff.buffer;
        }
        if (pinnedBuffers.size() == BUFFER_POOL_SIZE)
            throw new BufferAbortException();
        try {
            Buffer buff;
            long timestamp = System.currentTimeMillis();
            buff = bufferPool.pin(blk);
            if (buff == null) {
                waitingThreads.add(Thread.currentThread());
                while (buff == null && !waitingTooLong(timestamp)) {
                    bufferPool.wait(MAX_TIME);
                    if (waitingThreads.get(0).equals(Thread.currentThread()))
                        buff = bufferPool.pin(blk);
                }
                waitingThreads.remove(Thread.currentThread());
                bufferPool.notifyAll();
            }
            if (buff == null) {
                repin();
                buff = pin(blk);
            } else {
                pinnedBuffers.put(buff.block(), new PinnedBuffer(buff));
            }
            return buff;
        } catch (InterruptedException e) {
            throw new BufferAbortException();
        }
    }
}
```

Synchronize on the buffer pool (singleton)

```
public Buffer pin(BlockId blk) {
    synchronized (bufferPool) {
        PinnedBuffer pinnedBuff = pinnedBuffers.get(blk);
        if (pinnedBuff != null) {
            pinnedBuff.pinnedCount++;
            return pinnedBuff.buffer;
        }
        if (pinnedBuffers.size() == BUFFER_POOL_SIZE)
            throw new BufferAbortException();
        try {
            Buffer buff;
            long timestamp = System.currentTimeMillis();
            buff = bufferPool.pin(blk);
            if (buff == null) {
                waitingThreads.add(Thread.currentThread());
                while (buff == null && !waitingTooLong(timestamp)) {
                    bufferPool.wait(MAX_TIME);
                    if (waitingThreads.get(0).equals(Thread.currentThread()))
                        buff = bufferPool.pin(blk);
                }
                waitingThreads.remove(Thread.currentThread());
                bufferPool.notifyAll();
            }
            if (buff == null) {
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                buff = pin(blk);
            } else {
                pinnedBuffers.put(buff.block(), new PinnedBuffer(buff));
            }
            return buff;
        } catch (InterruptedException e) {
            throw new BufferAbortException();
        }
    }
}
```

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                while (buff == null && !waitingTooLong(timestamp)) {
                    bufferPool.wait(MAX_TIME);
                    if (waitingThreads.get(0).equals(Thread.currentThread()))
                        buff = bufferPool.pin(blk);
                }
                waitingThreads.remove(Thread.currentThread());
                bufferPool.notifyAll();
            }
            if (buff == null) {
                repin();
                buff = pin(blk);
            } else {
                pinnedBuffers.put(buff.block(), new PinnedBuffer(buff));
            }
            return buff;
        } catch (InterruptedException e) {
            throw new BufferAbortException();
        }
    }
}
```

Find the given block from the pinned buffers of this transaction

```java
public Buffer pin(BlockId blk) {
    synchronized (bufferPool) {
        PinnedBuffer pinnedBuff = pinnedBuffers.get(blk);
        if (pinnedBuff != null) {
            pinnedBuff.pinnedCount++;
            return pinnedBuff.buffer;
        }
        if (pinnedBuffers.size() == BUFFER_POOL_SIZE)
            throw new BufferAbortException();
        try {
            Buffer buff;
            long timestamp = System.currentTimeMillis();
            buff = bufferPool.pin(blk);
            if (buff == null) {
                waitingThreads.add(Thread.currentThread());
                while (buff == null && !waitingTooLong(timestamp)) {
                    bufferPool.wait(MAX_TIME);
                    if (waitingThreads.get(0).equals(Thread.currentThread()))
                        buff = bufferPool.pin(blk);
                }
                waitingThreads.remove(Thread.currentThread());
                bufferPool.notifyAll();
            }
            if (buff == null) {
                repin();
                buff = pin(blk);
            } else {
                pinnedBuffers.put(buff.block(), new PinnedBuffer(buff));
            }
            return buff;
        } catch (InterruptedException e) {
            throw new BufferAbortException();
        }
    }
}
```

Pins the requested block

Add the buffer to the pinned list of this transaction

```java
public Buffer pin(BlockId blk) {
    synchronized (bufferPool) {
        PinnedBuffer pinnedBuff = pinnedBuffers.get(blk);
        if (pinnedBuff != null) {
            pinnedBuff.pinnedCount++;
            return pinnedBuff.buffer;
        }
        if (pinnedBuffers.size() == BUFFER_POOL_SIZE)
            throw new BufferAbortException();
        try {
            Buffer buff;
            long timestamp = System.currentTimeMillis();
            buff = bufferPool.pin(blk);
            if (buff == null) {
                waitingThreads.add(Thread.currentThread());
                while (buff == null && !waitingTooLong(timestamp)) {
                    bufferPool.wait(MAX_TIME);
                    if (waitingThreads.get(0).equals(Thread.currentThread()))
                        buff = bufferPool.pin(blk);
                }
                waitingThreads.remove(Thread.currentThread());
                bufferPool.notifyAll();
            }
            if (buff == null) {
                repin();
                buff = pin(blk);
            } else {
                pinnedBuffers.put(buff.block(), new PinnedBuffer(buff));
            }
            return buff;
        } catch (InterruptedException e) {
            throw new BufferAbortException();
        }
    }
}
```

If there was not any available buffer, make the thread waiting

The thread in the head of the list can pin

Wake up other thread again

```java
public Buffer pin(BlockId blk) {
    synchronized (bufferPool) {
        PinnedBuffer pinnedBuff = pinnedBuffers.get(blk);
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        }
        if (pinnedBuffers.size() == BUFFER_POOL_SIZE)
            throw new BufferAbortException();
        try {
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            if (buff == null) {
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        } catch (InterruptedException e) {
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        }
    }
}
```

Waitting too long? There might be deadlock.
Re-pin all blocks

```java
public Buffer pin(BlockId blk) {
    synchronized (bufferPool) {
        PinnedBuffer pinnedBuff = pinnedBuffers.get(blk);
        if (pinnedBuff != null) {
            pinnedBuff.pinnedCount++;
            return pinnedBuff.buffer;
        }
        if (pinnedBuffers.size() == BUFFER_POOL_SIZE)
            throw new BufferAbortException();
        try {
            Buffer buff;
            long timestamp = System.currentTimeMillis();
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            if (buff == null) {
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                bufferPool.notifyAll();
            }
            if (buff == null) {
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            }
            return buff;
        } catch (InterruptedException e) {
            throw new BufferAbortException();
        }
    }
}
```

Self-deadlock: throw exception

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